

Microsoft® Train Simulator



open rails

Blender Exporter for MSTs and OPEN RAILS

Wayne Campbell & Pete Willard

Version .5, 2026-09-18

Contents

Microsoft Train Simulator/Open Rails exporter add-on for Blender	2
REVISION HISTORY	2
Conventions Used	4
Installation	6
Automatic Installation Steps	6
Manual Installation Steps	6
Examples	8
What's Changed With The 2.8 Upgrade	8
Overview	9
Optimizations	9
Arranging Parts And Lods In The Outliner	10
Using the Outliner Window	11
MSTS/ORTS Materials Panel	11
File Export Panel	14
Tutorial - Exporting A Crate	16
Optimizing Model Performance	20
Upgrading Models From Blender 2.79	24
Migration Steps	24
Advantages of Blender 2.8+	25
TODO Items	25
TODO for Future Updates	25
IDEAS for Future Updates	25

Microsoft Train Simulator and Open Rails exporter add-on for Blender

2026-09-18 5.2.5 Wayne Campbell & Pete Willard

COPYRIGHT (c) 2019 - 2026 by Wayne Campbell and Pete Willard, distributed under the terms of the GNU GPL V3

This program is free software: you can redistribute it and/or modify it under the terms of the GNU General Public License, Version 3, as published by the Free Software Foundation here. This program is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public License, included in the distribution package, for more details.

You are welcome to repost this package, with local language translations if needed, on your local community web site. Keep the package together with this document containing the copyright notice. Drop me an email so I can tell others where its available.

Microsoft Train Simulator/Open Rails exporter add-on for Blender

Blender 3.8+ Exporter for MSTs/OR, verified with Blender 5.2 LTS

- Version 5.2.5
- This is a script add-on for the Blender 3D program. Use it to create .S shape files for Microsoft Train Simulator or Open Rails.
- This version of the EXPORTER is for Blender 3.8 and newer. It has been updated for current Blender 5.x releases while retaining compatibility with older supported Blender versions where possible. Use V3.5 of the exporter for Blender versions before 2.8.

REVISION HISTORY

2026-09-18	Released V5.2.5 - pkw - Added OpenRails/MSTS export info sidecar reports
2026-09-10	Released V5.2.4 - pkw - Copy source texture file when no matching ACE/DDS texture exists
2026-09-10	Released V5.2.3 - pkw - Added optional copy of referenced ACE/DDS textures beside exported S files
2026-09-10	Released V5.2.2 - pkw - Added SNAP object-name keyword to retain export hierarchy without using animation keywords
2026-08-02	Released V5.2.1 - pkw - Blender 5.2 Action slot/f-curve compatibility and export cancel callback fix
2026-07-15	Released V5.2.0 - pkw - Blender 5.2 LTS compatibility verification
2026-06-21	Released V5.1.1 - pkw - Added more known animation part names
2026-06-19	Released V5.1.0 - pkw - Blender 5.x compatibility update and animation handling tweaks
2026-05-26	Released V5.0.0 - pkw - Blender 5.x shader node compatibility update
2025-11-06	Released V4.8.1 - pkw - Repackaged for easier installation in Blender
2025-10-08	Released V4.8 - pkw - Fix for Blender 4.5 Update for changes related to shader sockets
2025-01-19	Released V4.7 - pkw - Fix for the deprecated specular which is now IOR (Index of Refraction) in 4.x
	Note: It was causing the "recreateShaderNodes function to fail and cause the material nodes to all become disconnected.
2024-12-12	Released V4.6 - pkw - Fix for the deprecated to.mesh in 4.x
2024-04-03	Released V4.5 - pkw - Handling issues created by Blender 4.1 deprecating smoothing related APIs.
2023-07-13	Released V4.4 - pkw
	Added support for exporting texture references as DDS files instead of just ACE files using a checkbox.
	Like the ACE file export, it changes *ALL* texture references to DDS

when checked.

2019-07-23 Released V4.3
Improved instructions around linking a part to multiple LOD collections

2019-07-21 Fixed program error - no material slots, or empty material slot
Released V4.2 for testing
Added HierarchyOptimization
Added FastExport - improves speed 25 x
Restructured AddMesh, AddTriangle with MSTSMaterialsDetail structure to improve performance

2019-07-18 Indexed vertices for faster export (eg speedup by 3 times)
Released V4.1 to beta testers
Modified install package to work with 'Install from file..'
Fixed ShapeViewer crashes on animations(0)

2019-07-17 Released V4.0 to beta testers
Updated for Blender 2.8

2017-09-12 Released as V 3.5
added SubObjID to SubObjectHeader per report from 'Spike'
added support for Auto Smoothing per
<http://www.elvastower.com/forums/index.php?/topic/30491-blender-auto-smooth-and-msts-export/>

2016-01-11 Released as V 3.4
strip periods (.) from object names
fixes to LOD inheritance
remember RetainNames setting
file export dialog, default to .s extension

2016-01-10 Testing as V 3.3
Added RetainNames option
Fixed bounding sphere bug
Trimmed most numbers to 6 decimal places to reduce output file size

2016-01-10 Testing as V 3.2
Documented hidden option to use Railworks style LOD naming
Documented setting MAX values in Custom Properties
Fixed AttributeError: 'NoneType' object has no attribute
'game_settings'

2016-01-09 Testing as V 3.1
Added MipMapLODBias setting
Fixed bug, untextured faces - AttributeError: 'NoneType' object has no attribute 'name'

not needed for OR)
Implemented MSTs sub_object flags for sorting alpha blended faces (not needed for OR)
Implement MSTs sub_object flags for specularity (not needed for OR)
Added Alpha Sorted Transparency option to MSTs Material
Improved descriptions and naming of MSTs Material settings
Fixed MSTs Lighting value not being saved
Sped up iVertexAdd, iNormalAdd, etc, with spatial tree indexing, etc

2016-01-08 Testing as V 3.0

	Fix for WHEELS13, WHEELS23 Changed use_shadeless, from selecting Cruciform, now done with MSTS
Lighting	
	Added MSTS Material panel for lighting options, etc Added support for Railworks style LOD naming
2015-11-11	Released as V 2
	Changed prim_state labels to be compatible with Polymaster
2013-11-01	Initial Release as V 2.6.3

Conventions Used



Regular Note.



Pay attention to these.



You should know this.



With care, you can succeed.



Optional, but good to know.

Highlighting

BOLD

Italic

Source code

LMB

Left Mouse Button

MMB

Middle Mouse Button

RMB

Right Mouse Button

N-Panel

Number Panel, **N** Hotkey in Main 3D Window

[ENTER]

Sometimes you will see KEYBOARD entry look like this

Referenced footnotes appear at the **end** of Chapters

Web Links should be active and will open in your web browser if your PDF reader supports it.

Installation

Automatic Installation Steps

The current add-on ZIP from the GitHub Releases page can be installed directly from within Blender.

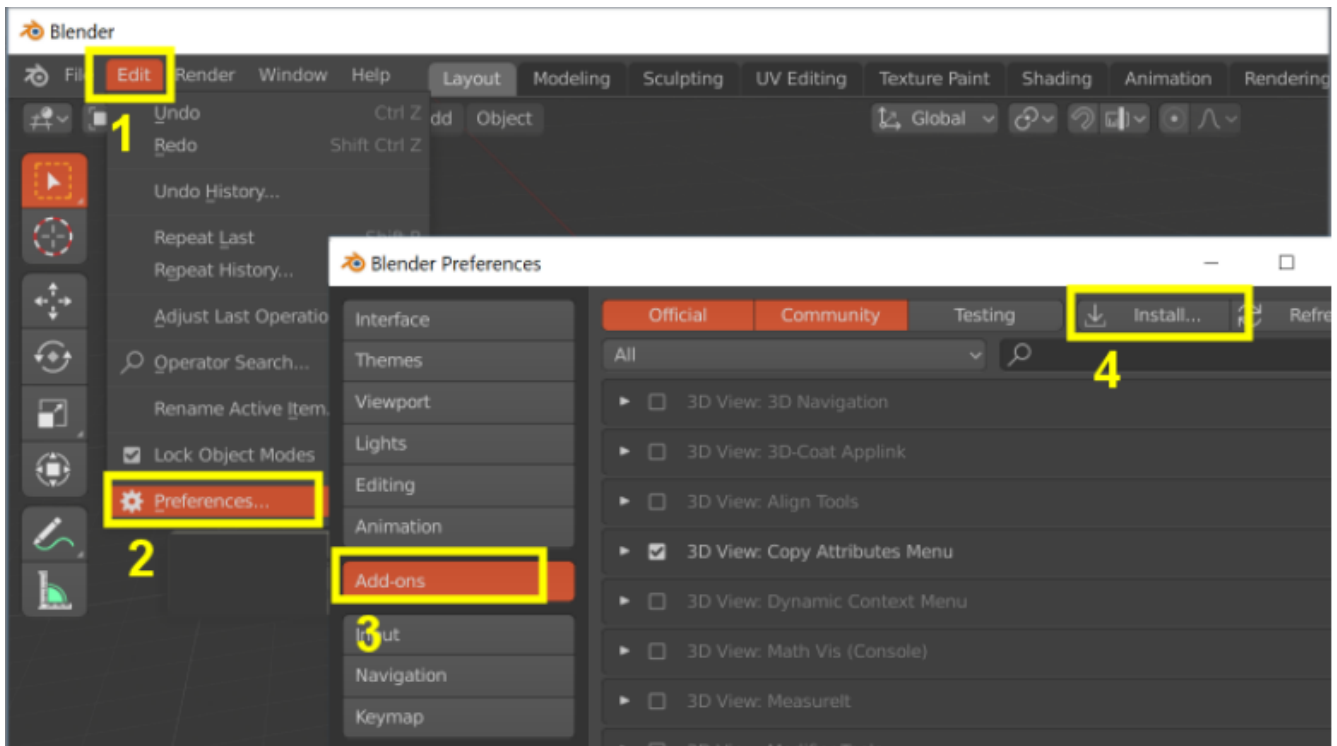
Install / update

- Blender → Edit → Preferences → Add-ons
- Choose the down-arrow in the upper right for the "Install from File" option
- Select the Exporter Add-On ZIP file, for example `MSTS_ORTS_Exporter_AddOn_v5.2.5.zip`.
- Enable OpenRails/MSTS Shape Exporter (Enable the CHECK mark, if not selected) .
- If updating from an older version, disable/remove the previous version of the add-on first to avoid duplicates.

As usual, to access the add-on, choose **File** → **Export** → **OpenRails/MSTS (.s)**.

Manual Installation Steps

Manual installation is only needed if Blender's install-from-ZIP workflow is not available. Open the downloaded add-on ZIP file and extract the `io_export_mstsexporter` folder to a convenient location.



To manually install the add-on, copy the extracted `io_export_mstsexporter` folder to your Blender "addons" folder, usually located at:



C:\Users\[username]\AppData\Roaming\Blender Foundation\Blender\[blender version]\scripts\addons

(The location is referring to your USERNAME and your current VERSION of Blender)

RESTART BLENDER if the Add-on does not show up.



The "Appdata" folder in a Windows machine is "hidden" by default, so you might need to go to file explorer options and adjust the settings so "hidden" folders are made visible.

Examples

Included in the package are:

Building\UnionStop

an example of a simple model that uses two different textures.

Loco\L1

a more complex example that includes animation and LOD levels.

Tutorial

contains texture file for use in the tutorial section of this document

What's Changed With The 2.8 Upgrade

The exporter now supports Blender 2.8's new Eevee rendering engine and materials. Unfortunately however Blender has discontinued support for Blender Internal materials and face texture methods. This means that any models originally made in 2.79 must be retextured if they are brought into Blender 2.8.

Another change relates to LODs and part naming. The updated exporter uses Blender's new 'collections' feature to organize the LODs of a model. We no longer have a model named 'MAIN'. Instead we have a collection named MAIN that contains all the parts we want to export.

More detail is provided later in the manual. Models originally made in 2.79 must be reorganized into collections when they are brought into Blender 2.8.

- The **DLEVEL**, **DMIN**, **DMAX** method of assigning LODS is no longer supported in the exporter.
- The Railworks style part naming is no longer supported in the exporter.
- The exporter no longer exports particle systems or bezier curves directly. These must be converted to meshes before export.
- The exporter is 4 times faster than the previous version (based on exporting L1 example file)

Overview

- look for the export script in Blender's File - Export menu.
- cloned objects and object modifiers are fully supported
- externally linked objects are fully supported
- if an object is not textured in Blender, the default texture, blank.ace, will be applied.

Optimizations

The script includes a number of optimizations for higher frame rates and better GPU and CPU usage including:

- Primitive consolidation which reduces batch calls by consolidating tris from different objects when they use the same material
- Large primitives, vertex_sets, and subobjects are split as needed to prevent them from exceeding MSTs's loading limits.
- Complex hierarchies are simplified and collapsed to prevent exceeding MSTs's hierarchy depth limits and to improve effectiveness of the primitive consolidation. However animated nodes are always retained.

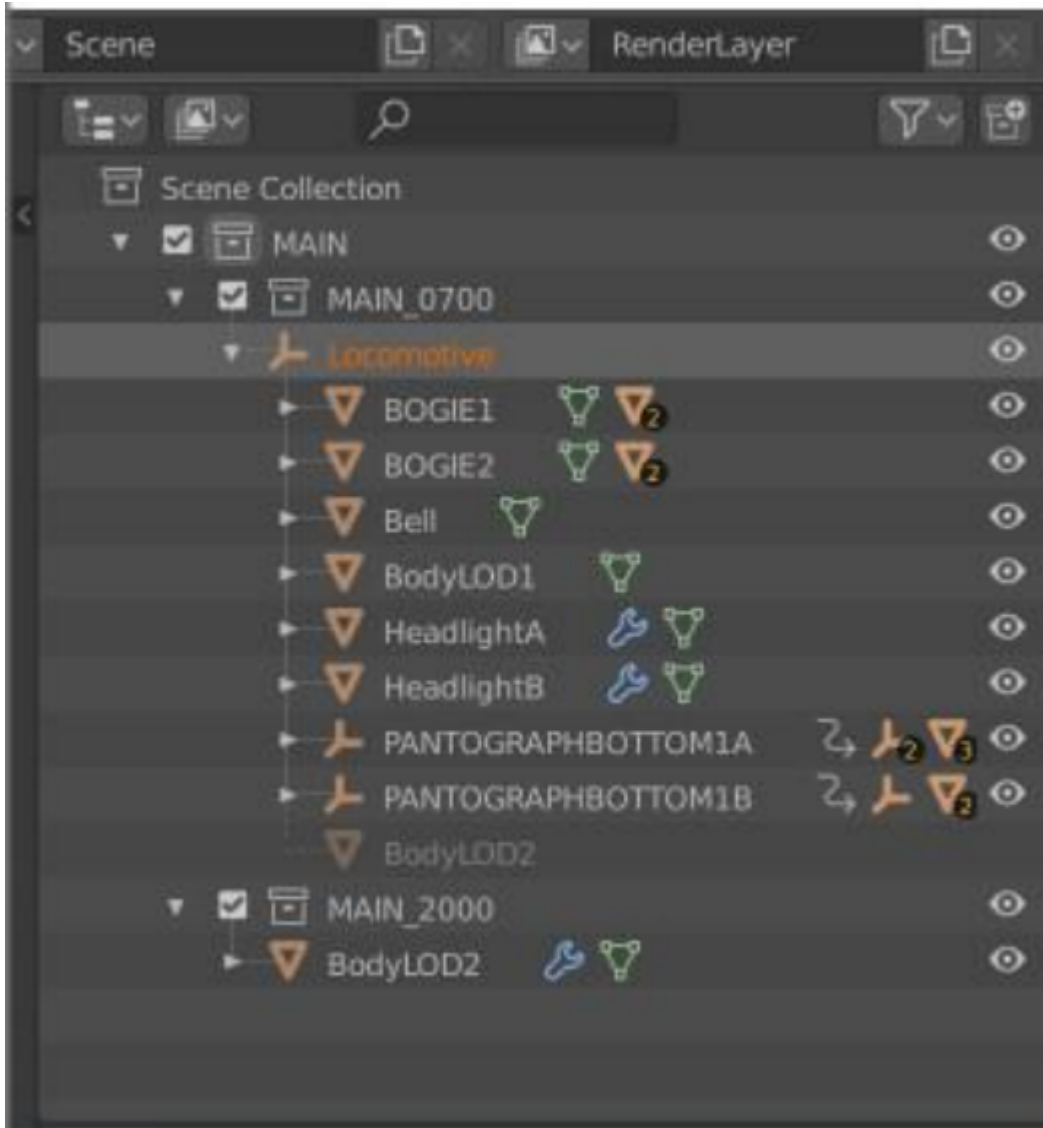
This includes ones where you have applied an animation in Blender, Open Rails-controlled parts, and the following commonly used retained names:

BOGIE1	BOGIE2	WHEELS	WHEELS	WHEELS	WHEELS	WHEELS	WHEELS	SNAP
		11	12	13	21	22	23	



You can disable this optimization and force the exporter to retain all part names with a checkbox option in the exporter window.

Arranging Parts And Lods In The Outliner



You must have a collection called **MAIN** in your scene. It must be attached directly to 'Scene Collection' **MAIN** contains one or more sub collection for each LOD, in this example, **MAIN_0700** and **MAIN_2000**. These LOD collections will contain all your parts.

You create parts and drag and drop them into whatever LOD you want. You can see in the example there is a BodyLOD1 in **MAIN_0700** and a lower resolution "BodyLOD2" in **MAIN_2000**.



"BodyLOD2" also appears grayed out in **MAIN_0700**. This shows you where it is in the Locomotive hierarchy, but grayed out since it isn't part of that LOD collection.

- You can control which LOD is visible in your viewport with the checkboxes beside each LOD collection. You can also create Render Layers that show one LOD or the other in a render.
- Use standard MSTs/Open Rails part names such as **BOGIE1**, **WHEELS11**, **WIPER**, **MIRROR**, and supported door/window/pantograph names for animated or simulator-controlled parts. (See the

"Optimizations" section for examples of recognized parts.) These key names **must** be in all capital letters to ensure they are retained in the exported shape file.

- Use **SNAP** as a neutral retained marker for target/helper shapes you want to preserve for use in TSRE or other utilities.
- Use lower case or mixed case names for all other parts.
- The exporter supports objects linked to other blend files (File, Link). With the limitation that animation in those linked files will not be exported.

Using the Outliner Window

- Dragging and dropping in the new outliner takes some practice. If you are moving a part from one LOD collection to another, just use the drag and drop method.
- If you are moving it to a new level in the hierarchy, drag and Shift drop. If you want a part to be seen further out, add it to more than one LOD using drag and [**Ctrl**] then drop.

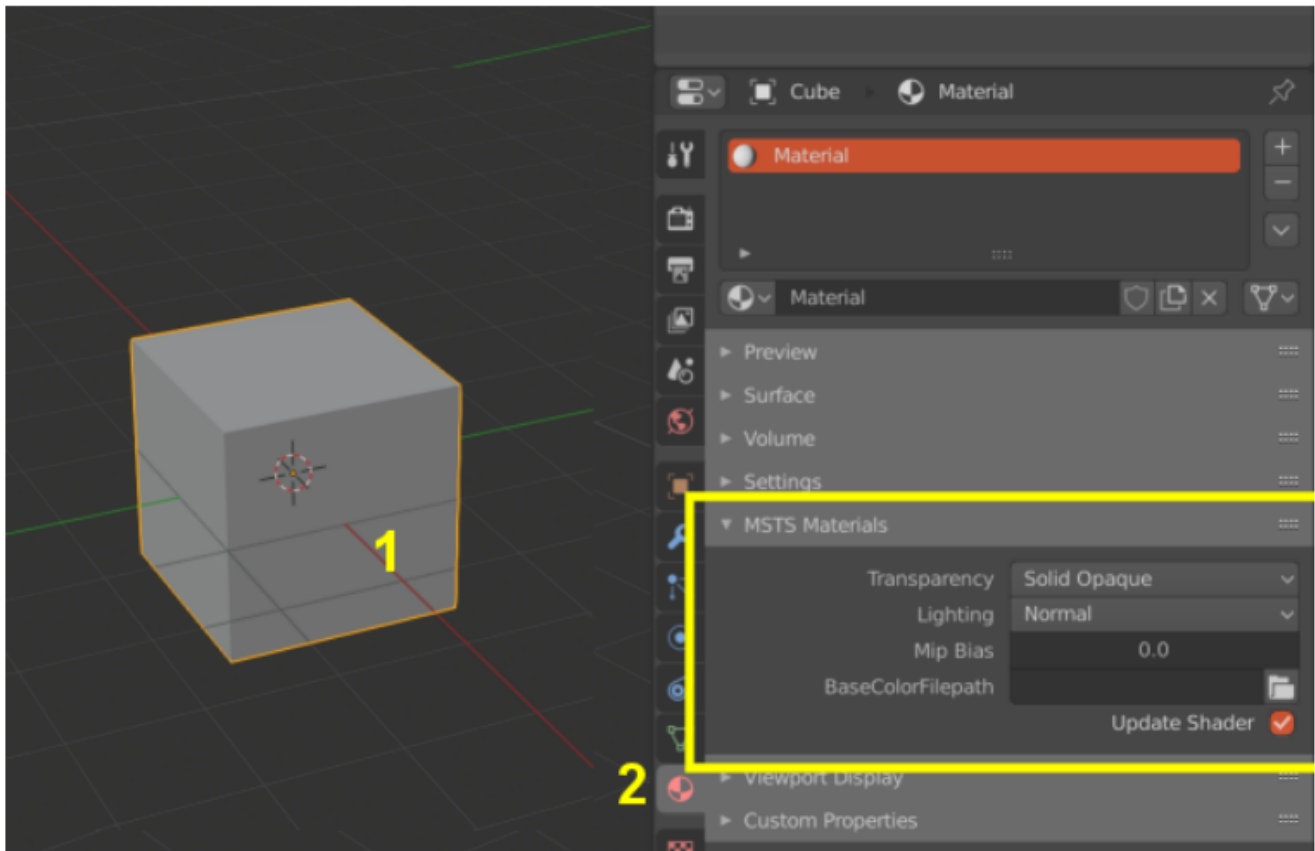


Instead of dragging and dropping, use the [**M**] key in the 3D viewport. Select an object, press [**M**], and choose the target LOD collections. Another option is to examine LOD collection assignment in the object tab of the properties panel.

[TIP] - if you use Blender primarily for MSTs/ORTS models, you can add a **MAIN** and an empty **LOD** collection to your initial startup file and then save it as the default startup with **File > Defaults > Save Startup File** to write this as your new startup settings.

MSTs/ORTS Materials Panel

To access the panel, select an object (1), choose the materials tab (2), find this panel near the bottom.



Item	Description
Transparency	Controls the effect of the texture's alpha channel
Solid Opaque	Alpha channel is ignored, it has no effect.
Transparency On/Off	Transparent if alpha value below a threshold
Alpha Blended	Alpha value blends from transparent up to opaque
Alpha Sorted	Alpha blending with scene depth sort
Lighting	Selects one of MSTs's special shading modes
Normal	Sun facing surfaces are lit and opposite facing are shaded
Specular 25	Strong specular highlight
Specular 750	Small specular highlight
Full Bright	Shaded surfaces appear lit
Half Bright	Shaded surfaces appear partly lit
Dark	Sun facing surfaces appear fully shaded
Cruciform	Indirect ambient lighting only
Emissive	Surfaces emit light at night
MipMapLODBias	Controls sharpness of the applied textures. (1)

Item	Description
BaseColorFilepath	Select the texture image (usually a .tga, .bmp, or .png file) to apply to the model.
Legacy Update Shader	Compatibility setting for older files. For current Blender versions, use the buttons below the material settings when you want to update Blender material display settings or rebuild the simple MSTs shader node setup. (2)

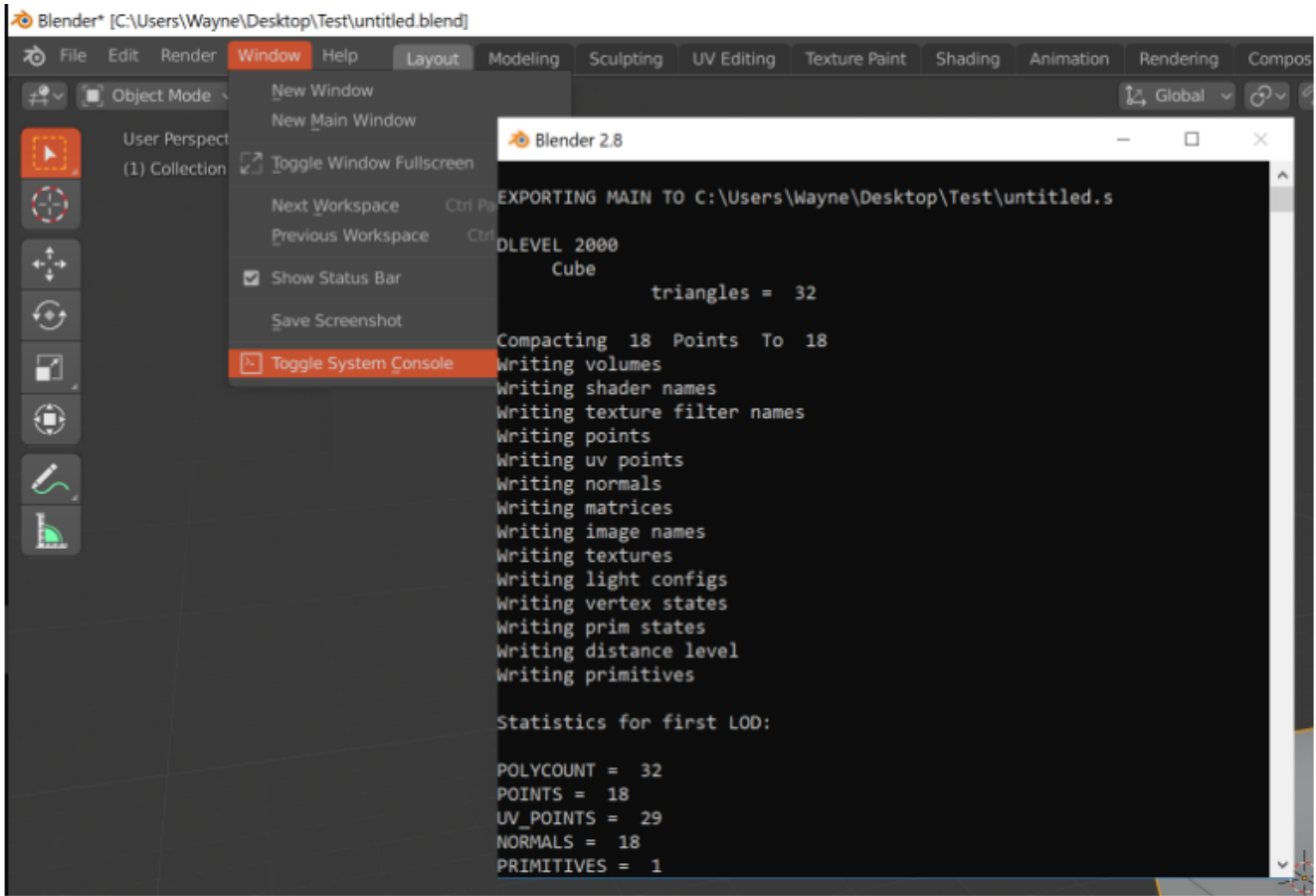
(1) Values range from -8 to +8. Negative values sharpen the image, but can cause excessive moire patterns. Positive values result in texturing blurring, and less moire.

(2) Apply MSTs Material Settings updates Blender's material display settings without replacing your shader nodes. Rebuild MSTs Shader Nodes replaces the material node tree with a simple MSTs-compatible shader setup. The Material name isn't used by MSTs or included in the exported .s file.

File Export Panel

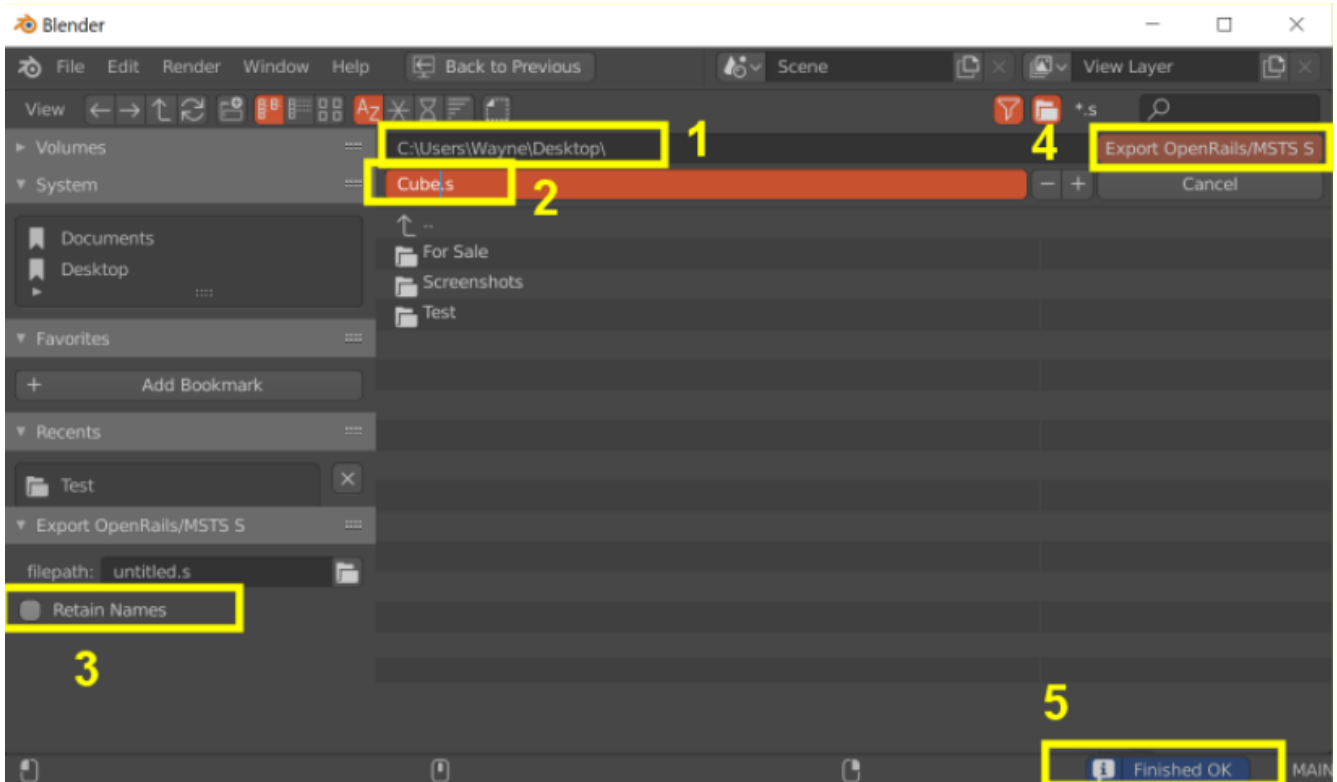


There is no progress bar (percent complete etc) so the program may appear to hang when exporting large files. Before you begin the export, toggle ON the system console (under the window menu) to see progress. There is a small counter shown at the cursor while the export is in progress.



ENSURE YOU ARE NOT IN EDIT MODE WHEN YOU EXPORT!

To reach the Export Panel, on the top menu bar choose File, Export, OpenRails/MSTS(.s)

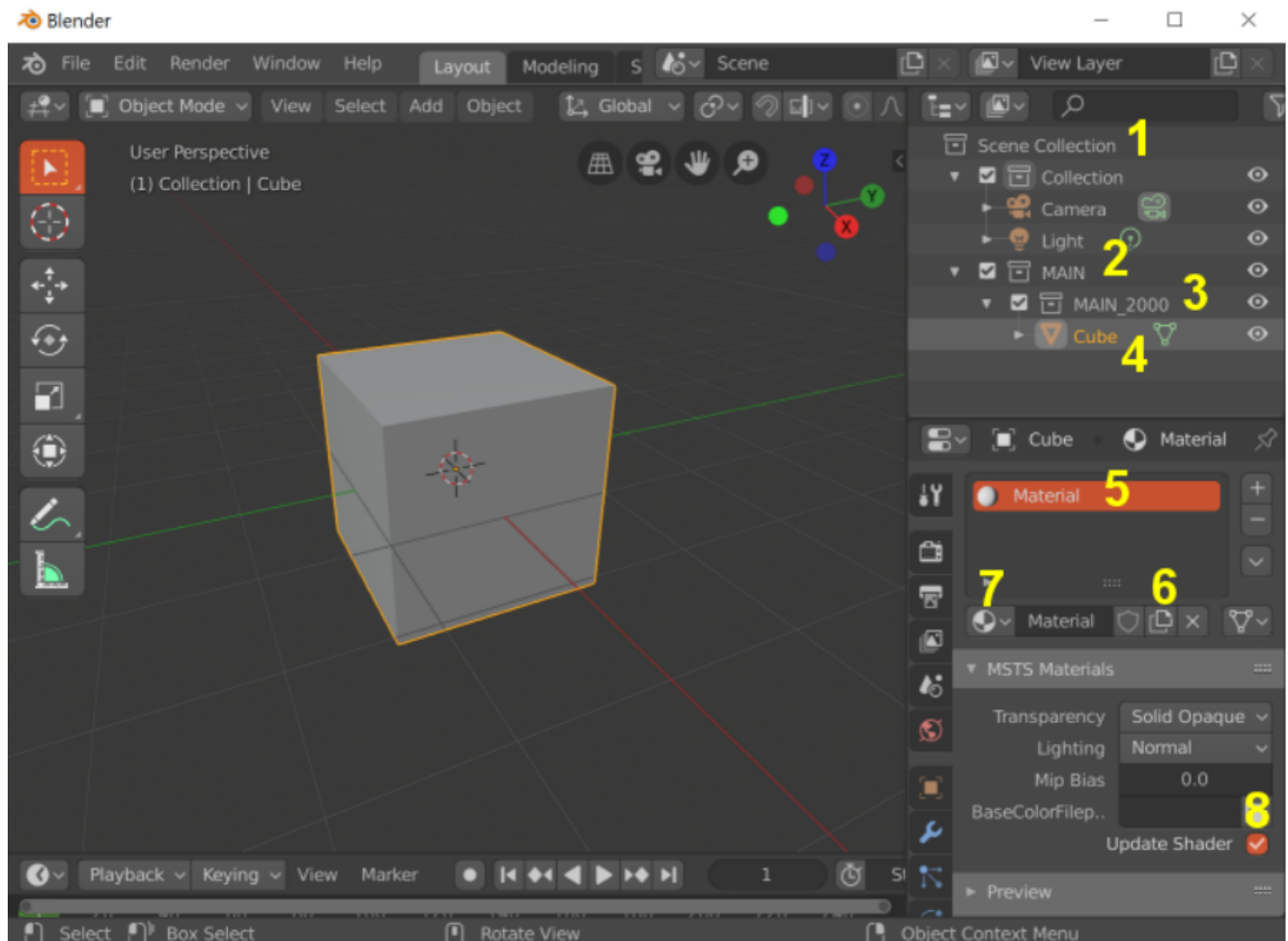


- Enter the directory that contains your shape files, or use the panel to the left to navigate to a folder.
- Enter the filename of the exported shape file.
- RETAIN NAMES - This setting disables most of the optimizations. All object names and hierarchy are retained in the shape file for users wanting better compatibility with external utilities such as Polymaster.
- USE DDS - This option, when checked, will export the .s file with textures referenced as DDS files instead of ACE
- COPY TEXTURES - This option copies texture files beside the exported .s file. If an ACE or DDS file matching the material base name exists, that runtime texture is copied. If no matching ACE/DDS file exists, the source texture used in Blender, such as BMP, TGA, or PNG, is copied instead. The exporter does not convert image formats.
- WRITE EXPORT INFO - This option writes `<shape>_export_info.json` and `<shape>_export_info.txt` beside the exported .s file. These sidecar files report the Open Rails/MSTS bounding box, volume sphere, exported image list, and LOD triangle/draw-call summary.
- Initiate the file export.
- Results will appear on the status panel at the bottom of the screen.

Tutorial - Exporting A Crate

This brief tutorial explains how to organize the parts in the outliner and then how to apply a texture to the cube. The final step is to export it to the MSTs .s shape format.

Open Blender to load the default cube.



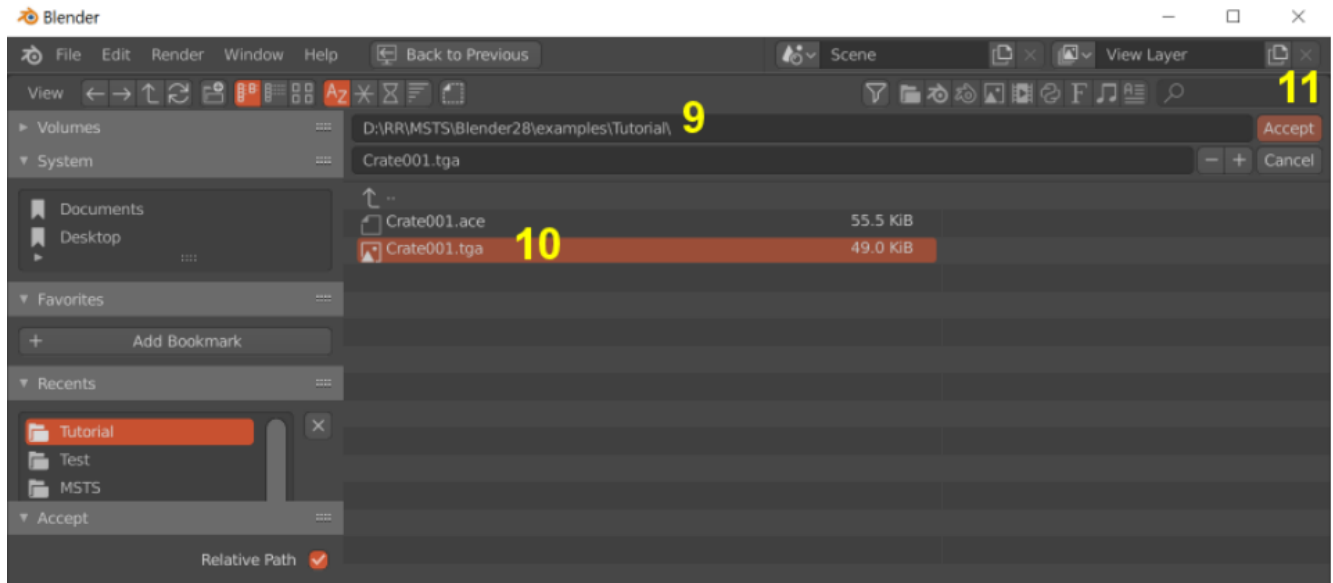
First set up the collection hierarchy.

Step Action

- 1 In the outliner, right click on 'Scene Collection' and create a new collection
- 2 Ctrl click on the new collection and rename it to MAIN. Right click on MAIN and attach a new collection to it
- 3 Ctrl click on this one and rename it to MAIN_2000. Left click (and release) to select the default Cube
- 4 Drag it to the **MAIN_2000** collection to assign it to that LOD

Next we will texture the cube to look like a crate.

- We will use the default material (5) that is already applied to the Cube,
- You could have created a new material with (6),
- or you could have selected from an existing material with (7).



Step Action

8

Click the file folder icon in the MSTS Materials panel to assign a texture to the default Material. The **Crate001.tga** texture is provided in the file download package included with this script.

9

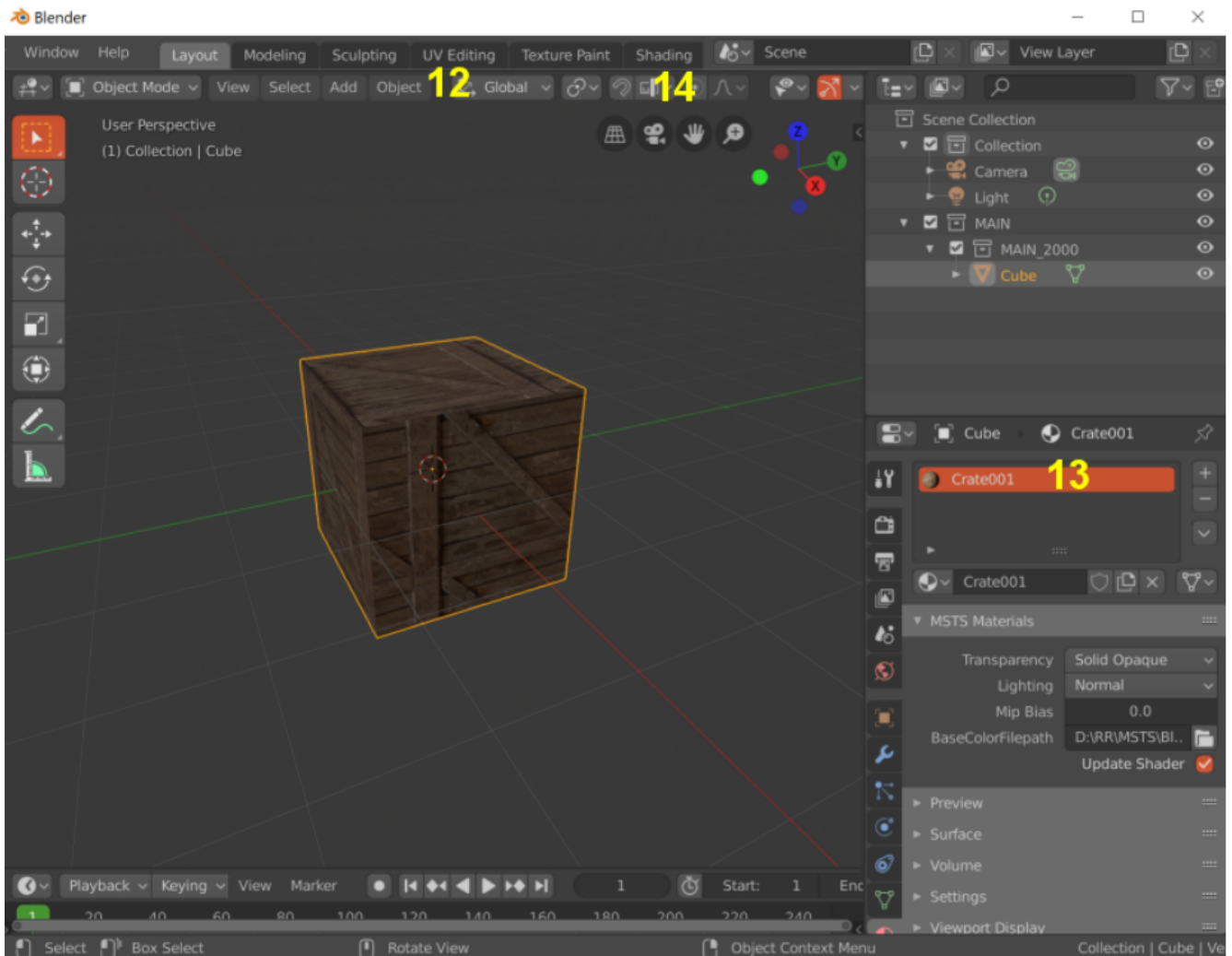
Enter a path to where you unzipped the Tutorial files or use the shortcuts on the left.

10

Choose the **Crate001.tga** texture file.

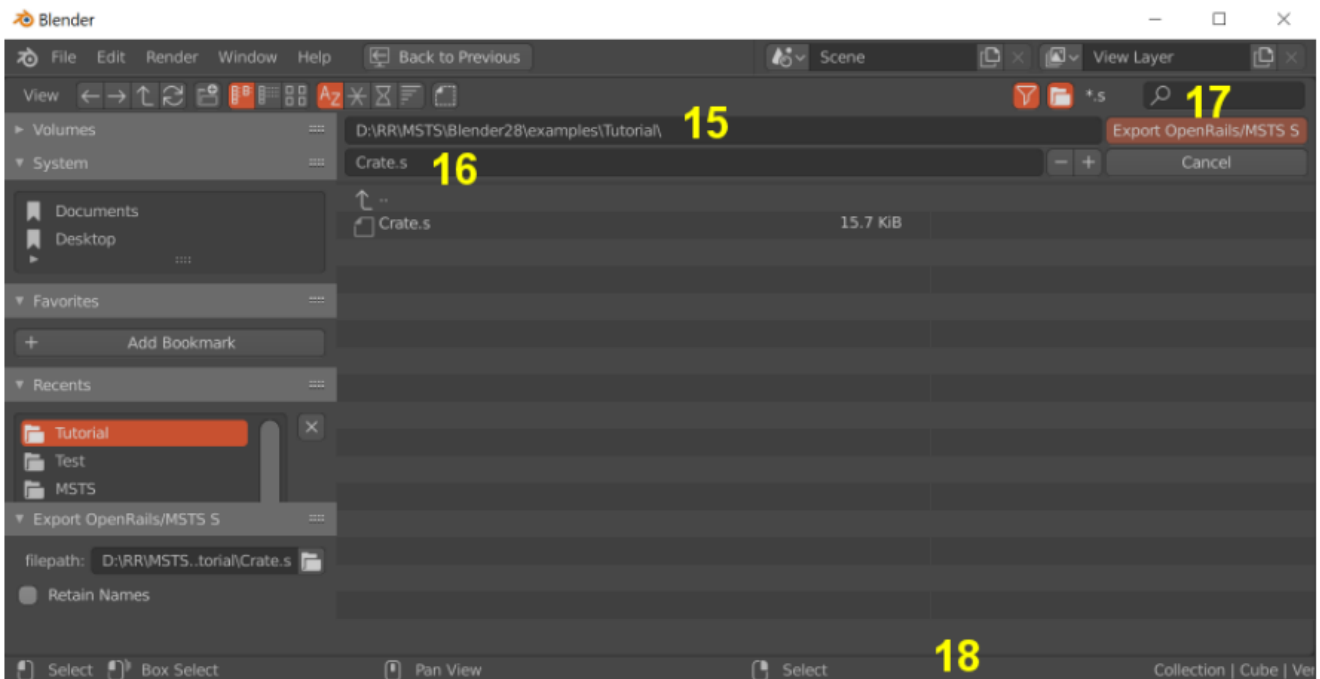
11

Press Accept.



Your cube will be textured.

- You can use your UV Editing (12) screen to adjust the texture mapping.
- Notice that the default Material was renamed to Crate001 (13). The MSTS Materials panel will do this so long as there isn't already a material with that name. Also, in the background, the panel has created a full Eevee shader.
- You can see it on the Shading (14) screen.
- Next, open the Export Panel from the top menu bar choose **File > Export > OpenRails/MSTS(.s)**



Step Action

- 15 Enter a path to the Tutorial folder or use the shortcuts on the left.
- 16 Export as Crate.S
- 17 Click on Export OpenRails/MSTS.
- 18 Look for 'Finished OK' in the status bar.



The exporter creates the .S file and, when enabled, sidecar export information and texture copies. Other route or rolling-stock definition files must still be created manually by other programs. For example:

- for scenery objects, you must create the .SD file and add an entry to the .REF file.
- for rolling stock you must create the .ENG or .WAG file
- you must convert the texture images to .ACE or .DDS files when the simulator or your workflow requires those runtime texture formats.

Optimizing Model Performance

The console window can help improve your model's performance. Modern graphics cards can draw millions of triangles per second using multiple GPU processors. However, the main limitation arises when setting up for drawing. Each time there is a change in texture image or material settings, the GPU pauses drawing to reconfigure and restart.

To optimize performance, it's important to minimize the number of Draw calls to the GPU. Below is an examination of the Blender log output for the sample engine. We can review it for useful details.

```
EXPORTING MAIN TO D:\git\Blender_MSTS_ORTS_Exporter\MSTSExporter
Examples\Loco\MSTS\lps-l1.s

DLEVEL 700
  HeadlightB
    triangles = 174
    Draw l1.ace CLIP NORMAL MipBias= 0.0 ②
  Bell
    triangles = 528 ③
  HeadlightA
    triangles = 174
  BodyLOD1
    triangles = 5010
    Draw chainlink.ace CLIP NORMAL MipBias= 0.0 ④
    Draw l1.ace ALPHA_SORT NORMAL MipBias= 0.0 ⑤
    Draw diamondplate.ace OPAQUE NORMAL MipBias= 0.0
    Draw l1.ace CLIP NORMAL MipBias= 0.0 ⑥
    Draw l1.ace CLIP NORMAL MipBias= 0.0 ⑦
  PANTOGRAPHBOTTOM1B
  Mesh.017
    triangles = 128
    Draw l1.ace CLIP NORMAL MipBias= 0.0 ⑧
  PANTOGRAPHMIDDLE1B
  Mesh.018
    triangles = 84
    Draw l1.ace CLIP NORMAL MipBias= 0.0
  PANTOGRAPHBOTTOM1A
  Mesh.014
    triangles = 128
    Draw l1.ace CLIP NORMAL MipBias= 0.0
  PANTOGRAPHMIDDLE1A
  Mesh.015
    triangles = 84
    Draw l1.ace CLIP NORMAL MipBias= 0.0
  PANTOGRAPHTOP1A
  Mesh.016
```

```

        triangles = 80
        Draw 11.ace CLIP NORMAL MipBias= 0.0
BOGIE2
        triangles = 196
        Draw 11.ace CLIP NORMAL MipBias= 0.0
WHEELS21
        triangles = 248
        Draw 11.ace CLIP NORMAL MipBias= 0.0
WHEELS22
        triangles = 248
        Draw 11.ace CLIP NORMAL MipBias= 0.0
BOGIE1
        triangles = 196
        Draw 11.ace CLIP NORMAL MipBias= 0.0
WHEELS11
        triangles = 248
        Draw 11.ace CLIP NORMAL MipBias= 0.0
WHEELS12
        triangles = 248
        Draw 11.ace CLIP NORMAL MipBias= 0.0
DLEVEL 2000
  BodyLOD2
    triangles = 48 ⑨
    Draw 11.ace OPAQUE NORMAL MipBias= 0.0

```

Compacting 9574 Points To 5265

```

Writing volumes
Writing shader names
Writing texture filter names
Writing points
Writing uv points
Writing normals
Writing matrices
Writing image names
Writing textures
Writing light configs
Writing vertex states
Writing prim states
Writing distance level
Writing primitives
Writing primitives
Writing primitives
Writing primitives
Writing primitives
Writing primitives
Writing primitives
Writing primitives

```

```
Writing primitives
Writing primitives
Writing primitives
Writing primitives
Writing primitives
Writing distance level
Writing primitives
```

```
LOD: 700
```

```
    Triangles = 7774 ①
```

```
    Draw Calls = 17
```

```
LOD: 2000
```

```
    Triangles = 48
```

```
    Draw Calls = 1
```

```
IMAGES:
```

```
    l1.ace
```

```
    diamondplate.ace
```

```
    chainlink.ace
```

```
EXPORT INFO:
```

```
    D:\git\Blender_MSTS_ORTS_Exporter\MSTSExporter Examples\Loco\MSTS\lps-  
l1_export_info.json
```

```
    D:\git\Blender_MSTS_ORTS_Exporter\MSTSExporter Examples\Loco\MSTS\lps-  
l1_export_info.txt
```

```
FINISHED OK
```

- ① At the end of the report, it tells you how many triangles are drawn for each LOD level and more importantly how many Draws were issued to the GPU. To understand what triggers these Draws you can look to the top section of the report
- ② The Headlight object triggers the first Draw using the **L1.ace** texture with the material settings shown
- ③ The Bell does not trigger a Draw because it uses the same texture and material as HeadlightB and so it was added to that Draw. Same for HeadlightA. We basically get these parts for free so it should be our objective to use the same texture and material on as many parts as possible
- ④ The BodyLOD1 object has some faces textured with 'chainlink.ace'. So this triggered a new Draw for that; same for 'diamondplate.ace'. BodyLOD1 also uses 'L1.ace' So far all of those faces are being added to the previous Draw for the headlight
- ⑤ Shows a new Draw set up for 'l1.ace'. This is because these faces have a material transparency set to ALPHA instead of CLIP. That triggers a new Draw for that setup. These are for the windows on the loco, so we will need to keep that extra draw
- ⑥ The next Draw uses the same texture and material as the one for Headlight. So you can assume it was triggered because the first Draw for Headlight was filled up. MSTS has a maximum number of triangles that it allows in a Draw (MSTS calls it a primitive)

- ⑦ So when we reach that limit, the exporter starts a new Draw. There is not much we can do to reduce these Draws except to trim out excess mesh triangles
- ⑧ Is for PANTOGRAPHBOTTOM1B. It uses the same texture and material settings as the previous one (7) but it can't be added to that Draw because this part is animated. Any animated node triggers a new Draw. That includes both ones animated in Blender, like PANTOGRAPHBOTTOM1B, as well as the automatically animated ones like BOGIE_x and WHEELS_{xx}. So its quite important to organize your animated parts to use only one texture and material settings per node , otherwise you could trigger multiple Draws for each animated part
- ⑨ The second LOD level is shown at here. It has just a single Draw. There are big performance benefits to organizing your distant LODs to use a single draw

If **Write Export Info** is enabled, the exporter also writes JSON and text sidecar files beside the exported shape. The JSON file is intended for tools such as OpenRails Engineer. It uses Open Rails/MSTS coordinates, where X is width/right, Y is height/up, and Z is length/forward. The text file is a quick human-readable bounding box report.

Upgrading Models From Blender 2.79

Version 2.8 was a 'breaking' change to Blender overall. In addition to fundamentally changing the user interface, this version did not just create an issue with MSTs files but also affects all files made in previous versions of Blender. Blender has introduced the new Eevee render engine and associated material system and discontinued the legacy 'Blender Internal' and 'Face Texture' system.

This means when you load an older version blend file that uses legacy materials, it will come into Blender 2.8 untextured and with empty material slots. However the UV mapping is retained at least which helps. For MSTs models you can use the new MSTs Materials panel to easily create the needed materials and shaders. If you have multiple textures on one mesh, you will use the 'Assign' tool to apply the new materials to the correct part of the mesh.

The other change relates to the method of assigning parts to LODs. The previous method, using custom Object Properties like **DLEVEL**, **DMIN**, **DMAX** etc has always been awkward to use. This seems like a good time to replace it with something better.

In the "Arranging Parts" section of this document there is an explanation of how the custom properties will be replaced with Blender 2.8's new 'collections' feature.

Migration Steps

In Blender 2.79 set Distance Level Selection to OFF and re-save file, this ensures all parts are in the visible state.

- In Blender 2.8, File Open with Load UI disabled
- Comes in without any texturing.
- Create **MAIN** collection
- Look at **MAIN** properties to determine **DLEVEL**'s needed for LOD Collections
- Create LOD Collections in **MAIN**
- Remove **DLEVELS** properties from **MAIN**
- Use **DMAX** and **DMIN** properties on each object to assign it to the correct LOD Collection
- Remove **DMAX** and **DMIN** Properties from all objects.
- Create and apply materials.
- UV Mapping should be OK
- Set Current Frame to 0 in Animation panel

Advantages of Blender 2.8+

Some of the advantages of collections over our previous method are:

- easy to assign LODs by dragging parts around or using the **[M]** key to assign collections
- easy to change distance level settings by editing LOD collection name, eg rename **MAIN_0700** to **MAIN_0500**
- good visibility of settings -all your LOD assignments show in the outline panel
- and the LOD assignments are also clearly shown in the object's 'Collection' panel
- you don't need the MSTs 'dlevel' control panel, choose LODs using the check boxes in the outliner
- future support for multiple LOD controllers, ie main has lods at 700 and 2000, but wheels have 100, 200 and 500.
- a part can be assigned to more than one LOD
- render in Cycles or Eevee, you can select which LOD appears in your render
- it uses standard Blender features - there's no hidden properties

TODO Items



To Maintainer: FINISH THIS SECTION WHEN BLENDER PROVIDES UPDATE PATH FOR 2.79 FILES. So far, nothing.

TODO for Future Updates

- add a progress bar (when available in the Blender API)
- document or remove hidden Normals override options (superceded by Blender's new Normals Modifier)

IDEAS for Future Updates

- add support for curves, particle systems, dupliverts
- support undocumented MSTs capability, eg
 1. Wrap, Clamp, Extend etc
 2. double sided faces
 3. bump mapping and environmental reflections
 4. AddATex, SubtractATex, etc and other undocumented shaders
 5. zBias

6. use of lod_control objects to improve LOD efficiency

- add optional texture conversion helpers
- expand sidecar export-info fields if OpenRails Engineer needs more data
- option to compress shape file
- options to generate .SD, .ENG, .WAG or .REF